A fixed-capability autonomous commercial loop under explicit simulated market conditions

AGI ALPHA Invention Flywheel Lab

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Abstract

We implemented and executed a bounded receivables-reconciliation service, a strategic repeated-market environment and an event-level financial auditor. The cohort supplied 19,000 authored synthetic records across 19 orders: 16 deliveries passed a separately implemented checker and 3 orders were rejected. Buyers, rival providers, free/conventional self-service and a supplier heuristic affect transactions; all customers, orders and payments are explicitly simulated. Across 168 prespecified four-arm seeded confirmation runs, the joint strategic-pricing/autonomous-onboarding arm offered $139.00 first-order and $121.62 repeat-order prices. In the declared qualification region, 13/18 runs supported repeated profitable delivery and continuation from prior unrestricted simulated earnings. This is a conditional model result over actually executed synthetic service artifacts, not observed commerce, population-level demand validation, recursive capability improvement or Blue Sky enablement.

1. Independent success definitions and evidence scope

Conclusion	**Machine verdict**	**Evidence scope**
profitable autonomous service	CONDITIONAL_SIMULATED_GO	SIMULATED_ECONOMICS_OVER_EXECUTED_SERVICE
simulated commercial flywheel	CONDITIONAL_SIMULATED_GO	SIMULATED
observed commercial flywheel	NOT_ESTABLISHED	OBSERVED
capability compounding flywheel	NOT_ESTABLISHED	CAPABILITY
blue sky enablement	NOT_ESTABLISHED	BLUE_SKY

Fixed-capability commercial success is independent of recursive invention. This campaign does not join the five conclusions with a global AND. The observed lane remains closed because no actual customer collection or exact provider bill was obtained. A successful simulated loop remains simulated in every downstream balance and investment; neither a simulation flag nor missing host billing automatically sets qualified simulated earnings to zero.

The later independent-commercial clarification in the supplied reference-v6 was read. Only its DOCX and extracted-text hashes are recorded here; its full contents are not published. Existing invention mechanisms, failed experiments and PR12-14 evidence are preserved separately. This ordinary service is neither relabeled an autonomous invention nor used to infer a fresh inherited-capability advantage.

2. Executed service production and acceptance

A bounded screen compared researched articles, tested software patches and receivables reconciliation. Reconciliation was selected for exact independently checkable outcomes. Orders contain USD integer-cent invoice and bank-payment records, customer identities, references and an explicit completeness obligation. Delivery produces a row-level JSON reconciliation and exception CSV. Referenced bundles and split payments match only at exact cents and consistent customer identity. Unreferenced payments require unique corresponding invoice and payment candidates. Overlapping exact reference groups remain quarantined.

The larger executed cohort contains 19 orders of 1,000 total source records each. Namespaced blocks synthesize diverse reference, split, bundled, unique-amount, partial, ambiguous and conflicting-reference patterns. Original completeness contracts are retained: valid exception schedules can be accepted only when the order allows unresolved findings; strict-completeness ambiguity is rejected. Unsupported CAD and a negative reversal are rejected at input validation; the strict ambiguous order receives a reconciliation that fails its completeness obligation. Thus 19,000 supplied records must not be relabeled 19,000 successfully reconciled records. The original smaller cohort and its exact source snapshots remain available; small source examples are not used to justify a four-hour human service estimate.

All inputs are authored public synthetic workloads. They are not client records or deidentified real books. Generation families and reused seed sets are disclosed; the larger execution is engineering replication, not a fresh population holdout. The unsafe-first-match output is a supplied adversarial counterexample, not a naturally observed frontier-model error rate. Its balanced arithmetic conceals an arbitrary ambiguous allocation, which the checker rejects. No tax filing, professional attestation, record modification, fraud certification or money movement is within this service contract.

The recorded production and checking CPU time for the cohort was 2.482850 seconds, with 2.483870 seconds wall time. These measurements exclude agent inference, code construction, orchestration and wider campaign work. They are not complete costs and are never converted into human work hours. Separate builder and checker roles share the execution host and model family; this is procedural separation, not external certification. Exact routed model/version, settings, token receipts and host billing were not exposed.

The added autonomous onboarding workflow creates clearly synthetic CSV exports from the frozen source records, then actually parses explicit contract metadata and invoice/payment tables into validated canonical order JSON. It supports the declared known CSV schema, exact headers, quoting/BOM, integer-cent units and explicit currency/reference checks. It does not claim an arbitrary financial-system integration or an actual customer export.

Across 19 orders, 17 imports reproduced the exact canonical source bytes, 2 inputs were rejected and 16 passed independent service acceptance. Total measured import/preparation plus production/requalification work was 2.578753 CPU seconds and 2.583986 wall seconds. These machine timings do not establish a monetary bill or zero customer attention. A malformed duplicate-header probe was rejected.

Item / payer	**Evidence**	**Timing / uncertainty**
Agent onboarding / seller	Actual CSV preparation, parsing and checks; machine work timed	First-customer setup budget $0.01-$2.00 is conditional, not exact billing; repeated imports still cost resources
Customer attention / buyer	Approval, switching, export and integration burden unobserved	Separate $0-$50 utility-friction range; not seller staff wages and not proved zero
External fees / named payer	No new paid external transaction in actual workflow	Default zero only for declared no-paid-provider case; stress when applicable

3. Strategic actors, payoffs and timing

The horizon is 8 cycles with 2 candidate deliveries per cycle, a finite cohort and bounded buyer budgets. No output can be sold twice within a run. Buyer participation is chosen from our service, a competent human-delivered alternative, an adapting rival, free/conventional self-service and no purchase. Buyers have heterogeneous latent values; arrivals, trust, switching costs, payment incentives and future reputation affect repeat choices. Rival decisions use noisy value proxies. Parameters are transparent conditional assumptions rather than observed willingness to pay. The 0.99 and 0.97 utility coefficients and trust levels are assumed ex ante reliability/value weights, not measured comparative accuracy estimates. Both completed-service routes are assessed against the same declared functional obligations; this experiment does not establish inferior human acceptance quality.

```
U_own = own_trust * 0.99 * value - own_price - onboarding + fast_delivery_premium
U_rival = rival_trust * rival_quality * value - rival_price - rival_switching + fast_delivery_premium
U_human = max(0, 0.97 * value - human_benchmark)
U_self = 0.99 * self_service_trust * value - self_service_cost + fast_delivery_premium
participate = arrival AND affordable AND U_own >= max(0, U_rival, U_human, U_self)
```

The competing provider searches a 30-point price/quality grid to maximize (price minus conditional unit cost) times estimated customer count. A reported zero residual verifies only maximization over that grid against noisy information. It is not a Nash equilibrium certificate, a continuous best response or a proof that no strategic deviation exists. The supplier uses a bounded scarcity pricing heuristic: the exogenous world term is multiplied by one plus 5% times batch utilization. Supplier terms and buyer preferences are structural assumptions; random seeds describe only variation within those assumptions.

Before each batch, the native exchange checks funding and commitments; actual deterministic service production and independent checking then run again on the selected source-bound orders. This sequencing prevents a claim that new earnings financed earlier qualification work. The actual host resource bill remains unknown: earned funds and resource payments in this demonstration are simulated. Provider investment and batch production precede buyer choice, so unsold inventory still costs resources. For a buyer who participates, consent, customer escrow funding and preparation precede release or refund; this is not order-specific consent before production. Delayed settlement, the contractual deadline, modeled revision charges, disputes, reputation updates, finite budgets and unsold work can prevent collection and repeat success. Market revision events are stochastic cost/risk draws, not separately executed corrected service artifacts; the actual artifact correction/rejection evidence is retained in the service cohort.

World	**Own / rival trust**	**Buyer friction original / corrected USD**	**Fast premium USD**	**Self-service USD / trust**
qualified_differentiation	0.99 / 0.79	25 / 5	25	30 / 0.85
moderate_differentiation	0.98 / 0.8	20 / 10	15	30 / 0.85
delayed_high_assurance	0.985 / 0.82	30 / 5	10	30 / 0.85
original_onboarding_barrier	0.99 / 0.79	25 / 25	0	30 / 0.85
commodity_ai_parity	0.97 / 0.99	0 / 0	25	5 / 0.99
zero_demand	0.99 / 0.9	0 / 0	0	5 / 0.99
correlated_adverse	0.72 / 0.96	0 / 0	0	5 / 0.99

The fast-delivery premium and lower onboarding assumptions define the targeted segment, not a measured discovery about actual buyers. The same premium is given to rival digital delivery and self-service. Fast conventional or self-service alternatives can therefore defeat the offer. The free-code alternative has zero software license price, while its declared user time/operating cost and execution trust remain uncertain model assumptions. Rival optimization uses own, human, self-service and abstention utilities with public switching assumptions, but private budgets and exact values remain uncertain. Its grid residual is conditional on that approximate information model.

4. Human comparison and premium pricing

The human benchmark is $139.00 for a four-hour manual-service scenario. BLS bookkeeping wages are loaded once using a published total-compensation-to-wages ratio; the rounded loaded rate is approximately $34.75 per hour. The four hours comprise assumed 0.5 hour first-customer human onboarding/setup plus 3.5 hours recurring human preparation, reconciliation, verification, corrections, delivery and support. The repeat-order loaded-labor benchmark is therefore $121.625 rather than adding onboarding again to $139. All human-side stages remain human work at human compensation. The task times are unmeasured scenario assumptions, not performance measurements against an accountant or strong import/reconciliation tools. The policy interval is $17.38 to $278.00. It is an assumed four-hour loaded-labor comparator covering the declared scope and obligations, with zero incremental nonlabor expense. No matched completed human workflow was measured; further enterprise costs are not invented or double counted. Fast competent conventional tools can reduce human time and erode the price anchor.

The initial offer applies a 5.0% declared discount to the human benchmark. Exploration tests $125.10, $132.05, $136.22, $139.00, $145.95, $166.80, $208.50 and evaluates the joint effects of price, discount and win probability on expected collected complete-cost profit and sustainable cash. The objective is neither maximum win rate nor the highest advertised price; the feasible premium-price grid and information assumptions bound any optimum claim. The selected first/repeat prices and their price-win-profit curves are reported below; adaptive candidate policies may set different multipliers on the first and repeat human anchors. Low AI cost is retained as potential margin rather than used as a cost-plus selling-price rule. A premium list price without buyer participation cannot qualify.

The human operation performs human work at human speed with loaded compensation. Agent-side research, production, verification, correction, acquisition, support, development and maintenance are machine workloads; no default human reviewer, salesperson or developer wage is inserted. No human performed per-job preparation, checking, correction, delivery or simulated transaction execution. The owner supplied the mandate and steering clarifications; these are disclosed inputs rather than hidden service staffing. No timed human service intervention was observed. Conventional and AI-native alternatives constrain willingness to pay; their technical parity does not automatically invalidate profitable autonomous service economics versus the human comparator.

Human stage	**First-order hours**	**Repeat-order hours**
onboarding	0.5	0
research	0.25	0.25
production	2.0	2.0
verification	0.5	0.5
correction	0.25	0.25
acquisition_ordering	0.25	0.25
support	0.25	0.25
Total	4.0	3.5

Original agent onboarding is contained within the recurring resource envelope; its separate zero line does not assert a free operation. The corrected arm adds $1.00 of explicitly conditional first-customer agent setup expense, with interval $0.01 to $2.00. Customer friction is a separate buyer utility term; external onboarding fee $0.00 is separate again. Setup is charged once when first consented technical acceptance occurs and remains completed after a refund; future orders are not guaranteed to amortize it.

The owner's four article, analysis, software-patch and accounting examples remain separate illustrative comparison fixtures. Their hypothetical budgets and lower reference selling prices are not measured service costs, current willingness to pay or the default premium-pricing rule. In particular, the accounting illustration is not the observed cost of this 1,000-record service. The present four-hour/$139 human scenario and conditional machine-resource envelopes are explicitly distinct assumptions.

5. Complete modeled costs and financial conservation

Quantity	Declared value / treatment
Opening simulated liquidity	$5,000.00; seed is not earnings
Restricted working capital	$100.00
Whole-campaign setup cost	$500.00; interval $200.00 to $2,500.00
Cost per cohort unit	$15.00; interval $5.00 to $30.00
Acquisition new / repeat	$4.00 / $1.00
Support allowance per scheduled delivery	$3.00
Payment charge	2.9% plus $0.30
Tax provision	20.0% conditional scenario; not a statement of applicable tax law

Setup includes campaign development, failed search, integration, acquisition design and maintenance. Per-unit envelopes include agent inference, tools, computation, storage, production, verification, corrections, rejected and unsold work, delivery and support obligations. Setup is charged in every arm, including NoBid. A completed horizon allocates all cohort and additional failed-attempt cost units through scheduled production batches, including unsold work. NoBid or an early stop avoids later unperformed batches; it does not erase incurred qualification/setup costs. The support allowance is charged per scheduled delivery, including unsold output, rather than only per collected sale. Source-exact token receipts are unavailable; the cost intervals are explicit conditional resource envelopes, not fabricated measured bills. A favorable result at the chosen point does not by itself prove profitability throughout the full declared interval.

The ledger adapts the existing native ProtectedExchange instead of redefining earned funds. The commerce006 loader checks the four vendor source hashes before import, and the paper manifest binds that loader, contracts, exchange implementation and preserved Apache-2.0 license. No other foundation repository is modified by this campaign. Monetary state uses integer simulated microUSD. The auditor reconstructs events and verifies source-bound cohort artifacts rather than accepting stored summary fields. Revenue, collection, refund, expenses, principal reserves, restricted custody, seed liquidity and available earned funds remain distinct; source order states record unsettled payment obligations. Commitment requires cash first; seed, deposits or restricted reserves cannot become earned capital. The full settled principal reserve and fixed per-order liability remain reserved through the modeled horizon, so accounting profitability and investment liquidity can differ.

Further acquisition and delivery capacity are funded only when the native unrestricted-earned gate admits the commitment and obligations remain covered. No capability-improved successor is required. The policy retains cash when incremental activity lacks support; capture, retained available earnings and actual simulated continuation expenditure are reported separately. Invention is an optional additional use of earnings, not a hidden condition for commercial qualification.

6. Four-arm pricing, onboarding and common-draw confirmation

Exploration seeds are [1, 2]; confirmation seeds are [23001001, 23001002, 23001003, 23001004, 23001005, 23001006]. The factorial arms are fixed 95% pricing/original onboarding, fixed 95% pricing/autonomous onboarding, strategic pricing/original onboarding, and both corrections. These are new controlled configurations under common final scenarios; fixed_original is not a rewrite or exact reexecution of the preserved earlier zero-sale experiment. Strategic policies maximize equally weighted exploration mean collected complete-cost profit subject to cash survival and no payable breaches, including NoBid as a cash-preserving alternative. Seller observations contain the declared prior value interval, public history and balances; private realized values, reservation prices and future draws are excluded. Pricing is frozen before confirmation; no target win-rate or perfect-GO gate chooses the price.

Common random inputs are keyed by seed, cycle, buyer and channel for values, arrivals, noisy rival information, revisions and disputes. This prevents conditional purchases from shifting later random draws between arms. Endogenous purchases, onboarding, trust, prices, balances and investment may diverge. The resulting paired

contrasts identify model-intervention effects under the stated policies and assumptions; they are not empirical causal effects of onboarding on actual customers. The revised development history is disclosed, and this finite model comparison makes no population-confidence or statistical-alpha claim.

Arm	First / repeat price	Expected complete profit	Settled / slot; cash survival	Mean repeat / earned cycles
fixed_original	$132.05 / $115.54	-$123.14	48.3% / 100.0%	1.06 / 1.83
fixed_autonomous_onboarding	$132.05 / $115.54	$367.99	76.4% / 100.0%	1.67 / 2.72
strategic_original	$125.10 / $109.46	$219.47	70.8% / 100.0%	1.56 / 2.22
strategic_autonomous_ onboarding	$139.00 / $121.62	$375.80	73.6% / 100.0%	1.61 / 3.06

Arm	Profit min / median / max	Mean available earned	Mean reinvestment	GO / runs
fixed_original	-$990.00 / $120.35 / $749.87	$268.46	$89.48	8/18
fixed_autonomous_onboarding	-$183.35 / $590.36 / $748.27	$394.76	$138.59	12/18
strategic_original	-$851.30 / $331.00 / $672.94	$312.46	$111.84	11/18
strategic_autonomous_ onboarding	-$807.66 / $662.57 / $825.21	$440.44	$155.04	13/18

The immutable legacy raw field win_probability is settled collections per scheduled opportunity slot, not conditional participation wins among arriving buyers. The reporting correction below separates arrival, selection, technical acceptance and collection without changing any financial result or verdict. All four arms share the same source-bound service quality and rejection rules: 16 qualified artifact contracts from 19 supplied orders. First and repeat price policies are the observed price-distribution support in this game; buyer state determines their mixture. Regional commercial qualification requires positive expected complete profit, all modeled runs surviving with no payable breach, and mean earned-funded cycles and repeat buyers meeting the declared thresholds. Individual run GO counts remain separately reported descriptive outcomes.

Arm	Arrival A/N	Conditional win W/A	Technical Q/W	Collection C/W	Settled C/N
fixed_original	94.10%	53.14%	100.00%	96.53%	48.26%
fixed_autonomous_onboarding	93.40%	84.76%	100.00%	96.49%	76.39%
strategic_original	93.40%	78.81%	100.00%	96.23%	70.83%
strategic_autonomous_ onboarding	93.40%	81.78%	100.00%	96.36%	73.61%

The combined arm has 288 scheduled slots, 269 arrivals, 220 participation wins, 220 native technical acceptances and 212 collections. Denominator-zero rates are undefined, not inferred zero. Native prepare/release counts and archived

decision outcomes are reconciled for all 168 confirmation runs. These are simulated buyer conversions; functional acceptance is an actual scoped synthetic-artifact replay, not a real-customer quality estimate. Shared underlying random draws remain matched even when endogenous purchase history changes repeat-arrival thresholds.

Reporting-v1 is preserved with its exact PDF and builder under history/reporting-v1. Its earlier win-rate display labels are corrected here to settled-per-slot conversion; prices, profits, verdicts and raw financial evidence are unchanged.

Integer price-policy labels are basis points of the applicable human anchor: 9500 means 95%. Adaptive labels encode first/repeat multipliers; None denotes NoBid rather than a free service. The displayed first/repeat prices describe each policy support, not an empirically observed willingness-to-pay distribution. All USD tables round to cents for display; replay uses exact integer simulated microUSD.

Exploration price response for strategic_original; selected policy 9000, finite-grid profitable-deviation residual $0.00. Zero residual is limited to this candidate grid, information model and sampled conditional prior, not Nash or globally optimal continuous pricing.

Price policy	**Expected complete profit**	**Settled / slot**	**Available cash / survival**
9000	$359.42	80.2%	$3,823.42 / 100.0%
9500	-$61.38	52.1%	$3,854.83 / 100.0%
9800	-$401.55	32.3%	$3,870.24 / 100.0%
10000	-$391.60	32.3%	$3,867.38 / 100.0%
10500	-$366.75	32.3%	$3,860.24 / 100.0%
12000	-$958.50	0.0%	$3,941.50 / 100.0%
15000	-$958.50	0.0%	$3,941.50 / 100.0%
first10500-repeat12000	-$302.97	32.3%	$3,841.92 / 100.0%
None	-$500.00	0.0%	$4,400.00 / 100.0%

Exploration price response for strategic_autonomous_onboarding; selected policy 10000, finite-grid profitable-deviation residual $0.00. Zero residual is limited to this candidate grid, information model and sampled conditional prior, not Nash or globally optimal continuous pricing.

Price policy	**Expected complete profit**	**Settled / slot**	**Available cash / survival**
9000	$481.77	88.5%	$3,796.94 / 100.0%
9500	$549.99	88.5%	$3,777.28 / 100.0%
9800	$590.93	88.5%	$3,765.48 / 100.0%
10000	$618.22	88.5%	$3,757.61 / 100.0%
10500	$160.63	60.4%	$3,799.88 / 100.0%
12000	-$292.71	32.3%	$3,838.28 / 100.0%
15000	-$958.50	0.0%	$3,941.50 / 100.0%
first10500-repeat12000	-$195.90	36.5%	$3,851.56 / 100.0%
None	-$500.00	0.0%	$4,400.00 / 100.0%

Paired intervention	All-world mean profit delta	Region mean profit delta	Delta min / median / max	Mean paid-job delta
onboarding_at_fixed_price	$210.48	$491.13	-$2.00 / $0.00 / $1,592.36	1.93
strategy_at_original_ onboarding	$308.43	$342.62	-$76.93 / $0.00 / $1,599.94	3.12
onboarding_at_strategic_price	-$94.60	$156.33	-$1,599.94 / $0.00 / $963.38	-1.38
both_vs_original	$213.83	$498.95	$0.00 / $0.00 / $1,664.57	1.74

The fixed-price onboarding contrast most directly isolates the modeled onboarding intervention. Strategic-original and strategic-autonomous arms may select different optimized prices; their contrast is a total reoptimized-policy effect, not a pure same-price onboarding effect. Paired controls preserve the same exogenous scenario inputs and budgets, while treatment-induced expense and transaction outcomes remain allowed to differ.

The following world-level distributions refer to the combined strategic/autonomous-onboarding arm. A conditional positive qualification does not imply success in every world or realization, and the faster-human, commodity and adverse alternatives remain material constraints.

World	GO / runs	Profit min / median / max	Paid jobs	Earned cycles
qualified_differentiation	6/6	$162.48 / $724.97 / $825.21	[9, 16]	[2, 5]
moderate_differentiation	3/6	-$807.66 / $419.99 / $699.77	[1, 15]	[0, 5]
delayed_high_assurance	4/6	-$141.37 / $662.57 / $674.57	[7, 15]	[0, 4]
original_onboarding_barrier	0/6	-$927.00 / -$927.00 / -$927.00	[0, 0]	[0, 0]
commodity_ai_parity	0/6	-$927.00 / -$927.00 / -$927.00	[0, 0]	[0, 0]
zero_demand	0/6	-$927.00 / -$927.00 / -$927.00	[0, 0]	[0, 0]
correlated_adverse	0/6	-$1,399.50 / -$1,399.50 / -$1,399.50	[0, 0]	[0, 0]

The qualification-region aggregate is 13/18 GO runs (72.2%), with modeled profit from -$807.66 to $825.21 and median $662.57. These distributions are conditional on the declared heuristic game and frozen cost point. They do not imply the corresponding frequency of real profitable businesses.

A reproducible median-profit qualification-region run is delayed_high_assurance, seed 23001001. Its terminal accounts below are simulated microUSD converted toUSD; opening liquidity is shown separately from available earned funds.

Terminal account	Median-run value
Seed liquidity	$5,000.00
Recognized revenue	$1,859.12
Collected cash inflow	$1,859.12
Complete modeled expenses	$1,196.56
Complete modeled profit	$662.57
Cash including seed	$5,662.57
Restricted principal/tail reserve	$1,862.12
Restricted custody	$0.00
Outstanding payables	$0.00
Total available liquidity	$3,700.44
Unrestricted earned funds	$662.57
Earned continuation expenditure	$221.00
Paid jobs / earned-funded cycles	15 / 4

A preserved earlier development run with $25 of explicitly unmeasured customer switching/attention friction and no fast-delivery premium collected 0 paid jobs and had modeled profit -$927.00. The $25 customer-friction assumption outweighed the $6.95 initial discount plus its modeled assurance difference; it was not an agent labor cost or human wage error. Actual CSV preparation does not measure buyer attention or prove that burden zero. This failed attempt informed the later bounded segment/onboarding assumptions; it is not omitted or promoted to fresh confirmation. The final protocol separately includes an original-onboarding-barrier world.

Prespecified sparse cost and human-time sensitivities follow. They rerun the first qualification world using separate fixed sensitivity seeds and the selected price multiplier; lower human time also lowers the price anchor. These points do not establish that the entire multidimensional cost interval is profitable.

Sensitivity case	Setup / unit / human USD	GO / runs	Profit min / median / max
cost_low	$200.00 / $5.00 / $139.00	3/3	$947.43 / $1,039.93 / $1,042.73
cost_region_edge	$700.00 / $20.00 / $139.00	3/3	$295.43 / $384.73 / $387.93
cost_high	$1,000.00 / $30.00 / $139.00	0/3	-$140.71 / -$34.09 / -$25.09
full_upper_cost	$2,500.00 / $30.00 / $139.00	0/3	-$1,640.71 / -$1,534.09 / -$1,525.09
human_half_hour	$500.00 / $15.00 / $17.38	0/3	-$695.51 / -$687.81 / -$686.31
human_two_hours	$500.00 / $15.00 / $69.50	0/3	-$108.46 / -$55.14 / -$53.64
human_eight_hours	$500.00 / $15.00 / $278.00	0/3	-$927.00 / -$927.00 / -$927.00
friction_0	$500.00 / $15.00 / $139.00	3/3	$539.43 / $630.73 / $631.93
friction_5	$500.00 / $15.00 / $139.00	3/3	$539.43 / $630.73 / $631.93
friction_25	$500.00 / $15.00 / $139.00	2/3	-$189.34 / $539.43 / $630.73

Sensitivity case	Setup / unit / human USD	GO / runs	Profit min / median / max
friction_50	$500.00 / $15.00 / $139.00	0/3	-$927.00 / -$927.00 / -$927.00

7. Robustness region, negative results and limitations

The declared qualification region comprises qualified_differentiation, moderate_differentiation, delayed_high_assurance. Its trust/assurance differentiation, buyer values and outside options are unmeasured structural assumptions. The service checker checks its narrow functional contract; it does not establish that customers grant that trust advantage or pay the modeled price. Synthetic well-specified inputs do not cover messy real exports, confidentiality operations, professional liability or unsupported accounting cases. Accepted exception-report service is not advertised as guaranteed fully resolved books.

Commodity AI parity, negligible demand and correlated adverse conditions are explicit failure controls. The correlated world combines weaker trust and arrival, stronger rival positioning, higher supplier cost, more revisions/disputes and delayed settlement. A positive result need not survive every adverse world, but failures must remain visible and may invalidate broad market claims. Conditional premiums depend on a segment that values the verified scope; no automatic near-AI-cost price cut is used to conceal failure.

The most informative next external experiment is a separately authorized blind evaluation on consented real invoice/payment exports, with actual task-time and complete-cost measurement followed by buyer choice at the premium price. That would discriminate the large unknowns: meaningful assurance differentiation, accepted exception scope, human-tool turnaround, conversion, repeat demand, corrections, collection and available cash. No outreach, new spending, credential change or real financial action is authorized by this campaign.

A separate provisional 25-service portfolio/pricing lane is linked in PR17 at source `be886e674f8b9bbce76410e5810b908c06a4300d`. Those hypothetical offers, win probabilities and profit illustrations are not executed service evidence from this campaign, validated buyer demand, observed revenue or proof of large-scale commercial capacity. Broader discount policies are possible future competing model controls; this frozen experiment is limited to 90-150% of the human anchor and its declared first/repeat adaptive candidate. No candidate was changed after the freeze to import a favorable portfolio result.

8. Reproduction, source identity and qualification

The initial recorded bounded allocation declares 120 wall minutes, 5400 CPU seconds, bounded storage and worker memory, zero new spending, zero paid provider calls, no real outreach and no real payment execution. These are allocation limits; unknown host inference usage is not manufactured into an exact resource receipt. Persistent improvement is bounded by this authority rather than an unlimited search for favorable seeds.

The disclosed recovery allocation retains 5400 prior conservative CPU charges and reserves at most 2400 additional seconds, for a cumulative 7800-second reservation ceiling inside the unchanged original 120-minute wall window. These conservative charges are not measured CPU consumption or a monetary bill; interrupted host usage remains unknown. No new spending, paid calls, outreach or real transactions are authorized, and the record forbids silently creating another allocation at the boundary.

Two failed operational attempts are preserved: transport loss left 52 complete exploratory records, and a later malformed SQLite database left 58 complete exploratory records; neither had executed a final confirmation case. Counts are overlapping cumulative inventories, not independent replications. The original freeze `68e4edb58962eac4110e4d9ff5e1845cd37ea5eb5450817c1431d3651715f2c1` and its exact old source closure remain archived. The physical corruption cause is unresolved, rather than claimed solved by a favorable retry.

The separately versioned I/O repair retains the pinned native financial methods and unchanged policy, costs, cohort and seeds. Active native SQL runs in memory; canonical action authority is fsynced and serialized checkpoints are integrity checked and atomically replaced. Receipt-path reporting is repaired separately. Only complete, checked

prior world records may be copied into a new recovery; malformed partial work is preserved and rerun. This is operational recovery and engineering reuse, not fresh independent commercial confirmation or a silent replacement of the original source identity.

Source refresh integrated main `69a4725d9c12589860670a551911bedbd65cba9e` with PR15 policy head `9e80181be5819bac7d4b472482db3e3604c75d46`. Historical operating-policy identity failures are repaired at the materialization boundary: immutable edition bytes and manifests are preserved separately while current operating documents can evolve. Benchmark discovery is wired into hosted qualification. Passing unrelated branch checks is not used to qualify the current source; final publication/merge/release checks remain source matched.

The final execution freeze SHA256 is `117ea809b09cb7bc8597bdcfb472e0df63a15d987dfedbc6844ef482d6c01ed5`. It binds policy, service inputs, the four-arm executable, independent audit, native exchange and bounded supervisor. Runtime versions are Python 3.12.14, SQLite 3.53.1 and cryptography 50.0.2. Frozen-input identity is checked before a worker starts; publication derivatives do not rewrite that operating-source freeze.

```
python run_commercial010.py --freeze-sha256 117ea809b09cb7bc8597bdcfb472e0df63a15d987dfedbc6844ef482d6c01ed5 --output
↪  runs/commercial010-replay
python scripts/run_onboarding010.py --output runs/onboarding010-replay
python scripts/run_service010.py --output runs/service010-replay
python -m unittest discover -s tests -v
python -m unittest discover -s benchmarks -v
python -m unittest discover -s verification -v
python scripts/build_commercial010_paper.py --results campaign/commercial-010/RESULT.json --policy campaign/commercial-010/POLICY.json
↪  --manifest campaign/commercial-010/service/executed-002/manifest.json --output docs/commercial010
```

The clean service-only reproduction copied producer, independent checker and runner into a temporary directory without Git history or network. A separate clean onboarding replay also reproduced all synthetic CSV bytes and 17 prepared order identities without Git or network; it does not supply real customer confirmation. All 19 input and produced-artifact hashes matched and output reuse was refused. The distributed release archive requires its own independent source-matched reproduction; this paper does not promote a local replay into released-archive qualification. SOURCE_MANIFEST.json binds input bytes, code and generated editions; exact tree identity is provided by publication metadata outside the paper to avoid self-referential commit claims.

9. Sources and provenance

- U.S. BLS May2025 occupational wage table
- U.S. BLS December2025 Employer Costs for Employee Compensation
- Xero bank-reconciliation workflow and automation guidance
- World Bank world GDP indicator
- ILOSTAT labor-income share discussion
- ILO SDG10.4.1 metadata
- Primary repository
- Alpha integration dependency issue4850
- Provisional separate portfolio/pricing PR17

The mandate reports world GDP of approximately $118.35 trillion and a 52.6% labor-income share for 2025. Their illustrative cross-source product is $62.2521 trillion annually, not an official combined measure of screen work or the AI-addressable market. The proposed $40 trillion opportunity remains a hypothesis. Neither scale figure calibrates orders, conversion or willingness to pay in this experiment. The ILO share already covers employee compensation and estimated self-employed labor income; no second benefit loading is applied to the global proxy. Source vintages and aggregation can differ. Occupation wages and broad compensation loading likewise do not measure this service task time. Source dates and applicability limits are retained in campaign calibration files.

```
Reference-v6 DOCX SHA256: 8916ec49efa7ed5c74d4a45e83731c88df108da6ea83977dfef654709457bdd2
Extracted-text SHA256: 03dbfd4a509407bbcc40c8485e4a98f12b0d919cce737fd0d7d4126027ced062
```